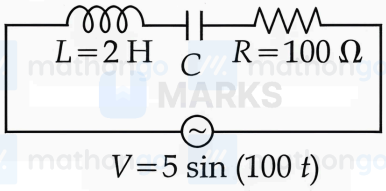
Q1. JEE Main 2026 (21 January Shift 2)

A capacitor C is first charged fully with potential difference of V_0 and disconnected from the battery. The charged capacitor is connected across an inductor having inductance L. In t s 25% of the initial energy in the capacitor is transferred to the inductor. The value of t is ____ s.

(1) $\frac{\pi\sqrt{LC}}{3}$ (2) $\frac{\pi\sqrt{LC}}{6}$ (3) $\frac{\pi\sqrt{LC}}{2}$ (4) $\pi\sqrt{\frac{LC}{2}}$

Q2. JEE Main 2026 (23 January Shift 1)

Using a variable frequency a.c. voltage source the maximum current measured in the given LCR circuit is 50 mA for $V = 5\sin(100t)$ The values of L and R are shown in the figure. The capacitance of the capacitor (C) used is ____ μF.



Q3. JEE Main 2026 (28 January Shift 2)

An inductor stores 16 J of magnetic field energy and dissipates 32 W of thermal energy due to its resistance when an a.c. current of 2 A (rms) and frequency 50 Hz flows through it. The ratio of inductive reactance to its resistance is ____. ($\pi = 3.14$)

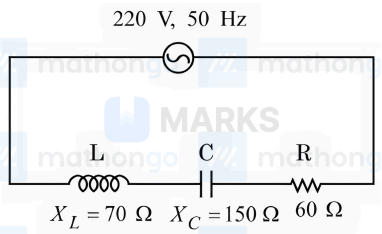
Q4. JEE Main 2026 (28 January Shift 1)

The electric current in the circuit is given as $i = i_o(t/T)$. The r.m.s current for the period $t = 0$ to $t = T$ is ____.

(1) i_o (2) $\frac{i_o}{\sqrt{6}}$ (3) $\frac{i_o}{\sqrt{3}}$ (4) $\frac{i_o}{\sqrt{2}}$

Q5. JEE Main 2026 (24 January Shift 1)

For the series LCR circuit connected with 220 V, 50 Hz a.c source as shown in the figure, the power factor is $\frac{\alpha}{10}$. The value of α is ____.



(1) 10 (2) 4 (3) 6 (4) 8

ANSWERS AND SOLUTIONS

1. (2) **2.** 50 **3.** 314 **4.** (3) **5.** (3)

1. **(2)** In an LC circuit, energy oscillates sinusoidally.
Initial energy in capacitor is $E_0 = \frac{1}{2}CV_0^2$.
When 25% is transferred to the inductor, 75% remains in the capacitor.
Since energy in capacitor varies as $E_C(t) = E_0 \cos^2(\omega t)$, we need $\cos^2(\omega t) = 0.75$.
This gives $\sin^2(\omega t) = 0.25$, so $\sin(\omega t) = 0.5$, yielding $\omega t = \frac{\pi}{6}$.
With $\omega = \frac{1}{\sqrt{LC}}$, we get $t = \frac{\pi\sqrt{LC}}{6}$ seconds.

2. **(50)** From $V = 5\sin(100t)$, angular frequency $\omega = 100$ rad/s
At resonance in LCR circuit: $X_L = X_C$, so $\omega L = \frac{1}{\omega C}$
Maximum current occurs at resonance: $I_{max} = \frac{V_{rms}}{R}$
$50 \times 10^{-3} = \frac{5/\sqrt{2}}{100}$ (verified at resonance)
From resonance condition: $100 \times 2 = \frac{1}{100 \times C}$
$200 = \frac{1}{100C}$
$C = \frac{1}{20000} = 5 \times 10^{-5}$ F = 50 μF

3. **(314)** Given: Magnetic field energy = 16 J, Thermal power dissipated = 32 W, $I_{rms} = 2$ A, $f = 50$ Hz.
From energy stored in inductor:
$\frac{1}{2}LI_{rms}^2 = 16$
$\frac{1}{2} \cdot L \cdot 4 = 16$
$L = 8$ H
From thermal power dissipated:
$I_{rms}^2 R = 32$
$4R = 32$
$R = 8\ \Omega$
Inductive reactance:
$X_L = \omega L = 2\pi f L = 2 \times 3.14 \times 50 \times 8 = 2512\ \Omega$
$\frac{X_L}{R} = \frac{2512}{8} = 314$

4. **(3)** The RMS current is calculated using $I_{rms} = \sqrt{\frac{1}{T}\int_0^T i^2 dt}$.
Given $i = i_0(t/T)$:

$$I_{rms} = \sqrt{\frac{1}{T}\int_0^T i_0^2\left(\frac{t}{T}\right)^2 dt} = \sqrt{\frac{i_0^2}{T^3}\int_0^T t^2 dt} = \sqrt{\frac{i_0^2}{T^3} \cdot \frac{T^3}{3}}$$

$$= \sqrt{\frac{i_0^2}{3}} = \frac{i_0}{\sqrt{3}}.$$

5. **(3)** Net reactance: $X = X_C - X_L = 150 - 70 = 80\Omega$.

Impedance: $Z = \sqrt{R^2 + X^2} = \sqrt{60^2 + 80^2} = \sqrt{10000} = 100\Omega$.

Power factor: $\cos\phi = R/Z = 60/100 = 0.6$.

Since power factor is $\alpha/10$, we have $0.6 = \alpha/10$, so $\alpha = 6$